BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 131 · Programming: Unseen Problems and Debugging

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does WHERE do in an SQL query?

Q2. When does a compiler translate source code?
A. When a key is pressed
B. Only after the program ends
C. Before the program runs
D. Only while each line runs

Q3. “A router's main job is executing spreadsheet formulas.” Correct this. Explain why.

Q4. A computer sends data to a printer on its LAN. “A switch sends every unicast frame to every port.” Correct this. How should the data reach the printer?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Understand the scenario — Identify input, processing, output and restrictions before coding.

Write a plan — Use simple steps, selections and loops before translating into code.

Trace unseen code — Record value changes and follow each condition accurately.

Debug and improve — Compare expected and actual behaviour, correct the cause and retest.

New learning

Understand the scenario: Identify input, processing, output and restrictions before coding.

Write a plan: Use simple steps, selections and loops before translating into code.

Trace unseen code: Record value changes and follow each condition accurately.

Debug and improve: Compare expected and actual behaviour, correct the cause and retest.

Worked example

An unknown program calculates price = quantity * unitPrice but prints unitPrice. Find the fault.

1. Identify expected output as total price.

2. Trace sample quantity=3, unitPrice=4; price=12.

3. The print statement uses the wrong variable.

4. Change to print(price), retest output 12.

We Do · Practise with guidance

Guided 1

A program should count 1 to 5 but stops at 4. Explain a fix.

Worked method: Test both first and last values.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the operator in a storage cost-per-GB calculation.

```
price = 48
capacityGB = 12
costPerGB = price * capacityGB
print(costPerGB)
```

iterate

Question 1

Explain why a trace is useful for unfamiliar code. [3 marks]

Question 2

Correct an algorithm that checks 1<=x<=10 using OR between two comparisons. [2 marks]

Question 3

State two steps to take after correcting a logic error. [2 marks]

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

A trace can help locate the first incorrect variable update.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It selects rows that meet a condition.

Q2. C — Before the program runs

Q3. It forwards traffic between different networks using IP addressing.

Q4. A switch uses MAC addresses to send data to the right connected device. It does not normally send each known unicast frame to every port.

I Do / We Do pages · Guided practice

Q1. Test both first and last values.

You Do pages · Code First

```
price = 48
capacityGB = 12
costPerGB = price / capacityGB
print(costPerGB)
```

Expected result: 4

You Do pages · Independent questions

Q1. It makes executed steps and changing variable values explicit, exposing incorrect assumptions or logic.

Q2. Use x>=1 AND x<=10.

Q3. Run/test with expected normal and boundary cases, and compare outcomes.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: